Dynamics Equations for Multiple Degree of Freedom Robots

The Lagrangian Mechanics can be used to define the equations of motion for a multiple degree of freedom robot. This approach involves defining the difference between the kinetic (K) and potential (P) energies of each link in the robot configuration i.e.

$$F_i = \frac{\partial}{\partial t}\left(\frac{\partial L}{\partial \dot{x}_i}\right) - \frac{\partial L}{\partial x_i}$$

$$T_i = \frac{\partial}{\partial t}\left(\frac{\partial L}{\partial \dot{\theta}_i}\right) - \frac{\partial L}{\partial \theta_i}$$

where

$$L = K - P$$

Kinetic Energy for a point on a link

The kinetic energy of a link is defined as:

$$K = \tfrac{1}{2} m V^2 + \tfrac{1}{2} I \omega^2$$

To define this relationship we must derive the velocity. This is achieved by considering the position of a point on the link, r_i, relative to its position on the fixed reference Frame, p_i:

$$p_i = {}^U T_i r_i$$

Differentiating this with respect to time gives the velocity:

$$V_i = \frac{dp_i}{dt} = \frac{d}{dt}\left({}^U T_i r_i\right)$$

This velocity can then be taken with respect to the velocities of the other joints:

$$V_i = \frac{d}{dt}\left({}^U T_i r_i\right) = \sum_{j=1}^{i} \left(\frac{\partial {}^U T_i}{\partial q_i} \frac{dq_j}{dt} \right) r_i$$

Derivatives of Transformations

The D-H transformation matrix for any joint *i* is formed in the following way

$$^{U}T_i = A_1 A_2 \cdots A_i$$

The derivative of this transformation matrix relative to joint *j* is found thus:

$$U_{ij} = \frac{\partial {}^{U}T_i}{\partial q_j} = \frac{\partial\left(A_1 A_2 \cdots A_j \cdots A_i\right)}{\partial q_j} = A_1 A_2 \cdots Q_j A_j \cdots A_i \qquad j \le i$$

Here *Q* is a constant matrix that satisfies the following relationship:

$$\frac{\partial A_j}{\partial q_j} = Q_j A_j$$

$$Q_j(prismatic) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad Q_j(revolute) = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The velocity for the point can be written with respect to the Joint velocities

$$V_i = \frac{d}{dt}\left({}^U T_i r_i\right) = \sum_{j=1}^{i}\left(\frac{\partial^U T_i}{\partial q_i}\frac{dq_j}{dt}\right) r_i = \sum_{j=1}^{i}\left(U_{ij}\frac{dq_j}{dt}\right) r_i$$

The related kinetic energy for the point is

$$dK_i = \tfrac{1}{2} V_i V_i^T dm_i = \tfrac{1}{2} Trace\left(V_i V_i^T\right) dm_i = \tfrac{1}{2}\left(\dot{x}_i^{\,2} + \dot{y}_i^{\,2} + \dot{z}_i^{\,2}\right) dm_i$$

Substituting for V_i gives

$$dK_i = \tfrac{1}{2} Trace\left(V_i V_i^T\right) dm_i = \tfrac{1}{2} Trace\left[\left(\sum_{p=1}^{i}\left(U_{ip}\frac{dq_p}{dt}\right) r_i\right) \cdot \left(\sum_{r=1}^{i}\left(U_{ir}\frac{dq_r}{dt}\right) r_i\right)^T\right] dm_i$$

The total kinetic energy in the link is

$$K_i = \int dK_i = \tfrac{1}{2} Trace\left[\sum_{p=1}^{i}\sum_{r=1}^{i} U_{ip}\left(\int r_i r_i^T dm_i\right) U_{ir}^{\,T} \frac{dq_p}{dt}\frac{dq_r}{dt}\right]$$

This can be rewritten as

$$K_i = \tfrac{1}{2}\sum_{p=1}^{i}\sum_{r=1}^{i} Trace\left(U_{ip} I_i U_{ir}{}^T\right)\frac{dq_p}{dt}\frac{dq_r}{dt}$$

Where

$$I_i = \begin{bmatrix} \left(-I_{xx}+I_{yy}+I_{zz}\right)/2 & I_{xy} & I_{xz} & m_i x_i \\ I_{xy} & \left(I_{xx}-I_{yy}+I_{zz}\right)/2 & I_{yz} & m_i y_i \\ I_{xz} & I_{yz} & \left(I_{xx}+I_{yy}-I_{zz}\right)/2 & m_i z_i \\ m_i x_i & m_i y_i & m_i z_i & m_i \end{bmatrix}$$

The total kinetic energy of all the links is

$$K_{links} = \frac{1}{2}\sum_{i=1}^{n}\sum_{p=1}^{i}\sum_{r=1}^{i} Trace\left(U_{ip} I_i U_{ir}{}^T\right)\frac{dq_p}{dt}\frac{dq_r}{dt}$$

The total kinetic energy of all the joints is

$$K_{actuators} = \frac{1}{2}\sum_{i=1}^{n} I_{i(actuators)}\left(\frac{dq_i}{dt}\right)^2$$

Therefore the total kinetic energy of the manipulator is

$$K = K_{links} + K_{actuators}$$

$$= \frac{1}{2}\sum_{i=1}^{n}\sum_{p=1}^{i}\sum_{r=1}^{i} Trace\left(U_{ip} I_i U_{ir}{}^T\right)\frac{dq_p}{dt}\frac{dq_r}{dt} + \frac{1}{2}\sum_{i=1}^{n} I_{i(actuators)}\left(\frac{dq_i}{dt}\right)^2$$

Potential Energy

The potential energy of an object is

$$P = mgh$$

It follows that the total potential energy of the links of a robot is

$$P = \sum_{i=1}^{n} P_i = \sum_{i=1}^{n} \left(- m_i \mathbf{g}^T \left({}^U T_i r_i \right) \right)$$

The Lagrangian

The Lagrangian can be written in the following form:

$$L = K - P$$

$$= \frac{1}{2}\sum_{i=1}^{n}\sum_{p=1}^{i}\sum_{r=1}^{i} Trace\left(U_{ip} I_i U_{ir}{}^{T}\right)\frac{dq_p}{dt}\frac{dq_r}{dt} + \frac{1}{2}\sum_{i=1}^{n} I_{i(actuators)}\left(\frac{dq_i}{dt}\right)^2$$

$$-\sum_{i=1}^{n}\left(-m_i \mathbf{g}^T \left({}^{U}T_i r_i\right)\right)$$

This gives the following equations of motion for link/joint i

$$T_i = \sum_{j=1}^{n} D_{ij} \frac{d^2 q_j}{dt^2} + I_{i(actuators)} \frac{d^2 q_i}{dt^2} + \sum_{j=1}^{n} \sum_{k=1}^{n} D_{ijk} \frac{dq_j}{dt} \frac{dq_k}{dt} + D_i$$

Where

$$D_{ij} = \sum_{p=\max(i,j)}^{n} Trace\left(U_{pj} I_p U_{pi}{}^T\right)$$

$$D_{ijk} = \sum_{p=\max(i,j)}^{n} Trace\left(U_{pjk} I_p U_{pik}{}^T\right)$$

$$D_i = \sum_{p=i}^{n} \left(-m_p \mathbf{g}^T U_{pi} r_p\right)$$

This can be written in full

$$
\begin{aligned}
T_i = {} & D_{i1}\ddot{\theta}_1 + D_{i2}\ddot{\theta}_2 + D_{i3}\ddot{\theta}_3 + D_{i4}\ddot{\theta}_4 + D_{i5}\ddot{\theta}_5 + D_{i6}\ddot{\theta}_6 + I_{i(actuators)}\ddot{\theta}_1 \\
& + D_{i11}\dot{\theta}_1{}^2 + D_{i22}\dot{\theta}_2{}^2 + D_{i33}\dot{\theta}_3{}^2 + D_{i44}\dot{\theta}_4{}^2 + D_{i55}\dot{\theta}_5{}^2 + D_{i66}\dot{\theta}_6{}^2 \\
& + D_{i12}\dot{\theta}_1\dot{\theta}_2 + D_{i13}\dot{\theta}_1\dot{\theta}_3 + D_{i14}\dot{\theta}_1\dot{\theta}_4 + D_{i15}\dot{\theta}_1\dot{\theta}_5 + D_{i16}\dot{\theta}_1\dot{\theta}_6 \\
& + D_{i21}\dot{\theta}_2\dot{\theta}_1 + D_{i23}\dot{\theta}_2\dot{\theta}_3 + D_{i24}\dot{\theta}_2\dot{\theta}_4 + D_{i25}\dot{\theta}_2\dot{\theta}_5 + D_{i26}\dot{\theta}_2\dot{\theta}_6 \\
& + D_{i31}\dot{\theta}_3\dot{\theta}_1 + D_{i32}\dot{\theta}_3\dot{\theta}_2 + D_{i34}\dot{\theta}_3\dot{\theta}_4 + D_{i35}\dot{\theta}_3\dot{\theta}_5 + D_{i36}\dot{\theta}_3\dot{\theta}_6 \\
& + D_{i41}\dot{\theta}_4\dot{\theta}_1 + D_{i42}\dot{\theta}_4\dot{\theta}_2 + D_{i43}\dot{\theta}_4\dot{\theta}_3 + D_{i45}\dot{\theta}_4\dot{\theta}_5 + D_{i46}\dot{\theta}_4\dot{\theta}_6 \\
& + D_{i51}\dot{\theta}_5\dot{\theta}_1 + D_{i52}\dot{\theta}_5\dot{\theta}_2 + D_{i53}\dot{\theta}_5\dot{\theta}_3 + D_{i54}\dot{\theta}_5\dot{\theta}_4 + D_{i56}\dot{\theta}_5\dot{\theta}_6 \\
& + D_{i61}\dot{\theta}_6\dot{\theta}_1 + D_{i62}\dot{\theta}_6\dot{\theta}_2 + D_{i63}\dot{\theta}_6\dot{\theta}_3 + D_{i64}\dot{\theta}_6\dot{\theta}_4 + D_{i65}\dot{\theta}_6\dot{\theta}_5 + D_i
\end{aligned}
$$

Static Forces

The forces/torques implemented at the end effector are represented as:

$$\left[{}^{H}F\right]=\left[f_x \;\; f_y \;\; f_z \;\; m_x \;\; m_y \;\; m_z\right]^T$$

These are made up of the forces/torques for the joints:

$$\left[T\right]=\left[T_1 \;\; T_2 \;\; T_3 \;\; T_4 \;\; T_5 \;\; T_6\right]^T$$

The total work done by the joints should equal the work done by the end effector:

$$\delta W=\left[{}^{H}F\right]^T\left[{}^{H}D\right]=\left[T\right]^T\left[D_\theta\right]$$

Considering the Jacobian relationship between the joints and the end effector, we get:

$$\left[{}^{H}F\right]^{T}\left[{}^{H}J\right]\left[D_{\theta}\right]=\left[T\right]^{T}\left[D_{\theta}\right]$$
$$\therefore\left[{}^{H}F\right]^{T}\left[{}^{H}J\right]=\left[T\right]^{T}$$

It follows that:

$$\left[T\right]=\left[{}^{H}J\right]^{T}\left[{}^{H}F\right]$$

Transformation of Forces between Frames

Forces can be transformed between frames in this way

$$[F]^T[D] = \left[{}^B F\right]^T \left[{}^B J\right][D]$$

$$\therefore [F]^T = \left[{}^B F\right]^T \left[{}^B J\right]$$

So that

$$[F] = \left[{}^B J\right]^T \left[{}^B F\right]$$

The elements of the frame forces can be calculated thus

$${}^B f_x = \mathbf{n} \bullet \mathbf{f}$$

$${}^B f_y = \mathbf{o} \bullet \mathbf{f}$$

$${}^B f_z = \mathbf{a} \bullet \mathbf{f}$$

$${}^B m_x = \mathbf{n} \bullet \left[(\mathbf{f} \times \mathbf{p}) + \mathbf{m}\right]$$

$${}^B m_y = \mathbf{o} \bullet \left[(\mathbf{f} \times \mathbf{p}) + \mathbf{m}\right]$$

$${}^B m_z = \mathbf{a} \bullet \left[(\mathbf{f} \times \mathbf{p}) + \mathbf{m}\right]$$